

Contoso Database Analysis : Unlocking Business Intelligence with SQL



\$ USD



£ GBP



F CHF



€ EUR



\$ CAD



\$ AUD



¥ JPY



元 CNY



kr SEK



\$ NZD



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INTRODUCTION

The ***Contoso database*** is a comprehensive sample dataset commonly used to demonstrate data analysis, reporting, and Business Intelligence (BI) techniques. It simulates a real-world retail business scenario, containing structured data on **customers**, **products**, **sales transactions**, **currency exchanges**, stores and **calendar dates**.

This presentation explores how meaningful insights can be derived from this data using **SQL queries**. By examining patterns across sales, customer behavior, product performance, and currency exchange trends, we demonstrate how Business Intelligence helps in making data-driven decisions that can improve business operations and strategy.

TOPIC DETAILS

- This presentation focuses on analyzing the **Contoso sample database**, which represents a fictional retail company's business operations. The database contains multiple interconnected tables including:
- **Calendar** – Provides date-related information used for time-based analysis.
- **CurrencyExchange** – Stores exchange rates between currencies over time.
- **Customer** – Contains customer details useful for segmentation and behavior analysis.
- **Products** – Lists product information such as names, categories, and prices.
- **Sales** – Records transactions linking customers, products, dates, and revenue.
- **Store** – Has all the records of products, customer data.

Through a series of SQL queries, this project demonstrates how to:

- Filter and retrieve meaningful records (e.g., weekend sales, Monday transactions).
- Join tables to combine data from multiple business areas.
- Group and summarize data to gain insights.
- Apply Business Intelligence techniques to support decision-making.
- The goal is to show how raw business data can be transformed into valuable insights using SQL — the foundation of Business Intelligence.

EER DIAGRAM

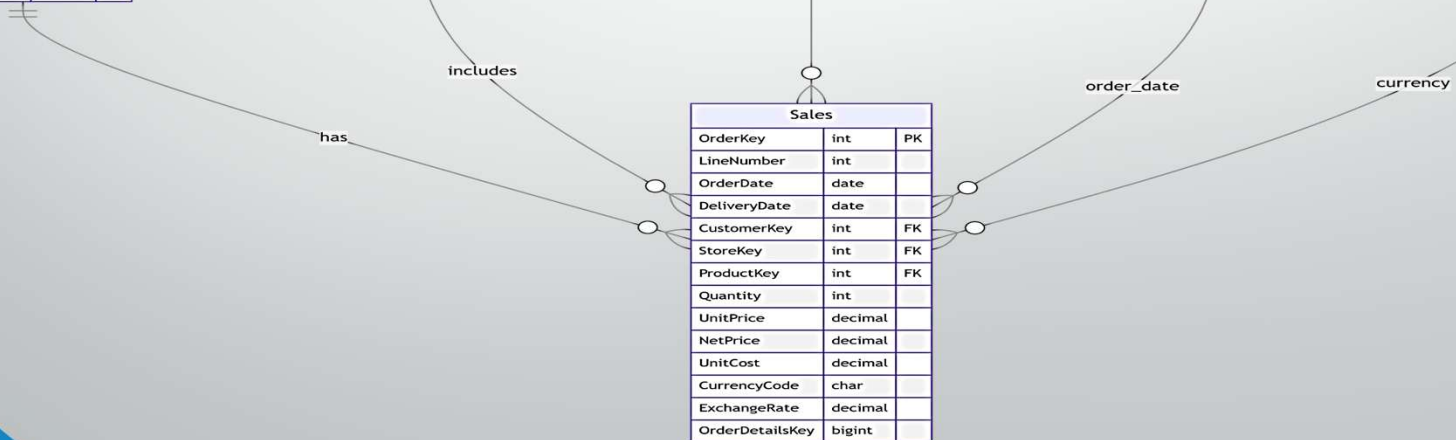
Customer		
CustomerKey	int	PK
GeoAreaKey	int	
StartDT	text	
EndDT	text	
Continent	text	
Gender	text	
Title	text	
GivenName	text	
MiddleInitial	text	
Surname	text	
StreetAddress	text	
City	text	
State	text	
StateFull	text	
ZipCode	int	
Country	text	
CountryFull	text	
Birthday	text	
Age	int	
Occupation	text	
Company	text	
Vehicle	text	
Latitude	double	
Longitude	double	

Products		
ProductKey	int	PK
ProductCode	int	
ProductName	varchar	
Manufacturer	varchar	
Brand	varchar	
Color	varchar	
WeightUnit	varchar	
Weight	decimal	
Cost	decimal	
Price	decimal	
CategoryKey	int	
CategoryName	varchar	
SubCategoryKey	int	
SubCategoryName	text	

Store		
StoreKey	int	PK
StoreCode	int	
GeoAreaKey	int	
CountryCode	varchar	
CountryName	varchar	
State	varchar	
OpenDate	varchar	
CloseDate	varchar	
Description	int	
SquareMeters	varchar	
Status	varchar	

Calendar		
date	date	
datekey	char	PK
year	char	
qyear	char	
YearQuarterNumber	char	
quarter	char	
yearmonth	char	
yearmonthshort	char	
YearMonthNumber	char	
Month	varchar	
MonthShort	char	
MonthNumber	smallint	
DayofWeek	varchar	
DayofWeekShort	char	
DayofWeekNumber	smallint	
WorkingDay	smallint	
WorkingDayNumber	int	

CurrencyExchange	
Date	text
FromCurrency	text
ToCurrency	text
Exchange	double



1. Get all records in the 'calendar' where 'WorkingDay' = 1.

SELECT * **FROM** calendar **WHERE** WorkingDay = 1;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	date	datekey	year	qyear	YearQuarterNumber	quarter	yearmonth	yearmonthshort	YearMonthNumber	Month	MonthShort	MonthNumber	DayofWeek	DayofWeekShort	DayofWeekNumber	WorkingDay	
▶	2015-01-02	20150102	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Friday	Fri	6	1	1
	2015-01-05	20150105	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Monday	Mon	2	1	2
	2015-01-06	20150106	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Tuesday	Tue	3	1	3
	2015-01-07	20150107	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Wednesday	Wed	4	1	4

1 06:33:53 SELECT * FROM calendar WHERE WorkingDay = 1

2509 row(s) returned

0.203 sec / 0.015 sec

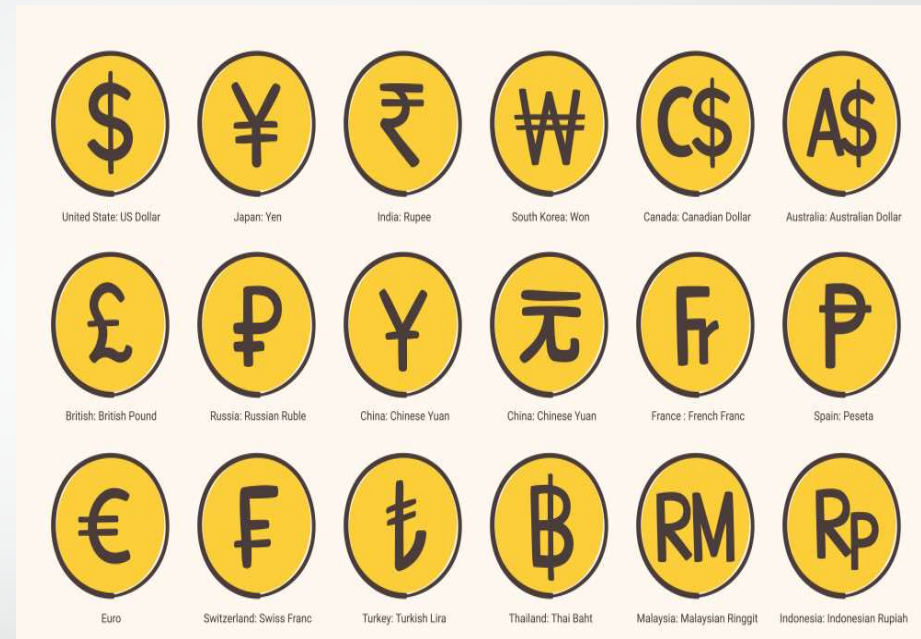
Result Grid



2. How to retrieve exchange rates where `FromCurrency` is not equal to `ToCurrency`.

```
SELECT * FROM currencyexchange  
WHERE FromCurrency != ToCurrency ;
```

	Date	FromCurrency	ToCurrency	Exchange
►	2015-01-01	AUD	CAD	0.94834
	2015-01-01	AUD	EUR	0.67435
	2015-01-01	AUD	GBP	0.52525
	2015-01-01	AUD	USD	0.81873
	2015-01-01	CAD	AUD	1.05447



3.How can we get the number of working days per month ?

```
SELECT Year, Month, MonthNumber, COUNT(*) AS WorkingDays
FROM calendar
WHERE WorkingDay = 'Yes'
GROUP BY Year, Month, MonthNumber
ORDER BY Year, MonthNumber;
```

	Year	Month	MonthNumber	WorkingDays
►	2015	January	1	11
	2015	February	2	9
	2015	March	3	9
	2015	April	4	8
	2015	May	5	11



5 07:05:43 SELECT Year, Month, MonthNumber, COUNT(*) AS WorkingDays FROM calendar WHERE WorkingD... 120 row(s) returned

0.188 sec / 0.000 sec

4. How to find the sum of 'WorkingDayNumber' per 'MonthShort'.

```
SELECT year, monthshort, SUM(WorkingDayNumber) AS workingdayNo_sum  
FROM calendar  
GROUP BY Year, MonthShort;
```

	year	monthshort	workingdayNo_sum
▶	2015	Jan	307
	2015	Feb	837
	2015	Mar	1562
	2015	Apr	2167
	2015	May	2907

23 07:25:28 select year, monthshort, SUM(WorkingDayNumber) AS workingdayNo_sum from calendar GROUP BY Year, Month... 120 row(s) returned

0.172 sec / 0.000 sec

5. How to Find the average exchange rate for each 'FromCurrency'

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
188
189 -- 2. Find the average exchange rate for each 'FromCurrency'.
190
191 • SELECT fromcurrency, AVG(Exchange) AS Exchange_Avg
192   from currencyexchange
193   GROUP BY fromcurrency;
194
195
```

Result Grid

fromcurrency	Exchange_Avg
AUD	0.767839486449489
CAD	0.8188279770052106
EUR	1.2005868283602374
GBP	1.4137208486175878
USD	1.0746754032302206

Result 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	06:50:01	SELECT fromcurrency, AVG(Exchange) AS Exchange_Avg from currencyexchange GROUP BY fromcurrency	5 row(s) returned	1.422 sec / 0.000 sec

6. How to get number of working days

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1 x

Don't Limit

```
-- 4. Get the number of working days per month.
SELECT
    Year,
    Month,
    MonthNumber,
    COUNT(*) AS WorkingDays
FROM calendar
WHERE WorkingDay = 'Yes'
GROUP BY Year, Month, MonthNumber
ORDER BY Year, MonthNumber;
```

Result Grid

	Year	Month	MonthNumber	WorkingDays
▶	2015	January	1	11
	2015	February	2	9
	2015	March	3	9
	2015	April	4	8
	2015	May	5	11
	2015	June	6	8
	2015	July	7	9
	2015	August	8	10
	2015	September	9	0

Result 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	06:58:43	SELECT Davofweek. COUNT(*) AS Total Days from calendar group by DavofWeek	7 row(s) returned	0.453 sec / 0.000 sec

7. How to get exchange rates by date and list how many conversions

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

218

219 -- 5. Group exchange rates by date and list how many conversions occurred per day.

220

221 • select Date, count(*) as conversion_day

222 from currencyexchange

223 GROUP BY Date;

224

Result Grid

Date	conversion_day
2015-01-01	25
2015-01-02	25
2015-01-03	25
2015-01-04	25
2015-01-05	25
2015-01-06	25
2015-01-07	25
2015-01-08	25
2015-01-09	25
2015-01-10	25
2015-01-11	25
2015-01-12	25
2015-01-13	25

Result 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	07:01:15	SELECT	Year, Month, MonthNumber, COUNT(*) AS WorkingDays FROM calendar WHERE WorkingD...	120 row(s) returned 0.453 sec / 0.000 sec

Query Completed

8. Find max exchange rate per `FromCurrency`.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
-- 6. Find max exchange rate per `FromCurrency`.
select FromCurrency, MAX(exchange) AS Exchange_Max
from currencyexchange
GROUP BY FromCurrency;
```

Result Grid

FromCurrency	Exchange_Max
AUD	1.03223
CAD	1.19832
EUR	1.8635
GBP	2.20541
USD	1.7253

Result 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	07:04:38	select Date, count(*) as conversion_day from currencyexchange GROUP BY Date	3653 row(s) returned	0.969 sec / 0.015 sec

Query Completed

9. Group by 'quarter' and calculate total working days per quarter.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
-- 7. Group by 'quarter' and calculate total working days per quarter.
select year, quarter, COUNT(*) AS WorkingDays_Quarter
from calendar
WHERE WorkingDay = 1
GROUP BY year, quarter;
```

Result Grid

	year	quarter	WorkingDays_Quarter
▶	2015	Q1	61
	2015	Q2	86
	2015	Q3	82
	2015	Q4	22
	2016	Q1	62
	2016	Q2	84
	2016	Q3	84
	2016	Q4	21
	2017	Q1	62
	2017	Q2	84
	2017	Q3	84
	2017	Q4	20
	2018	Q1	62

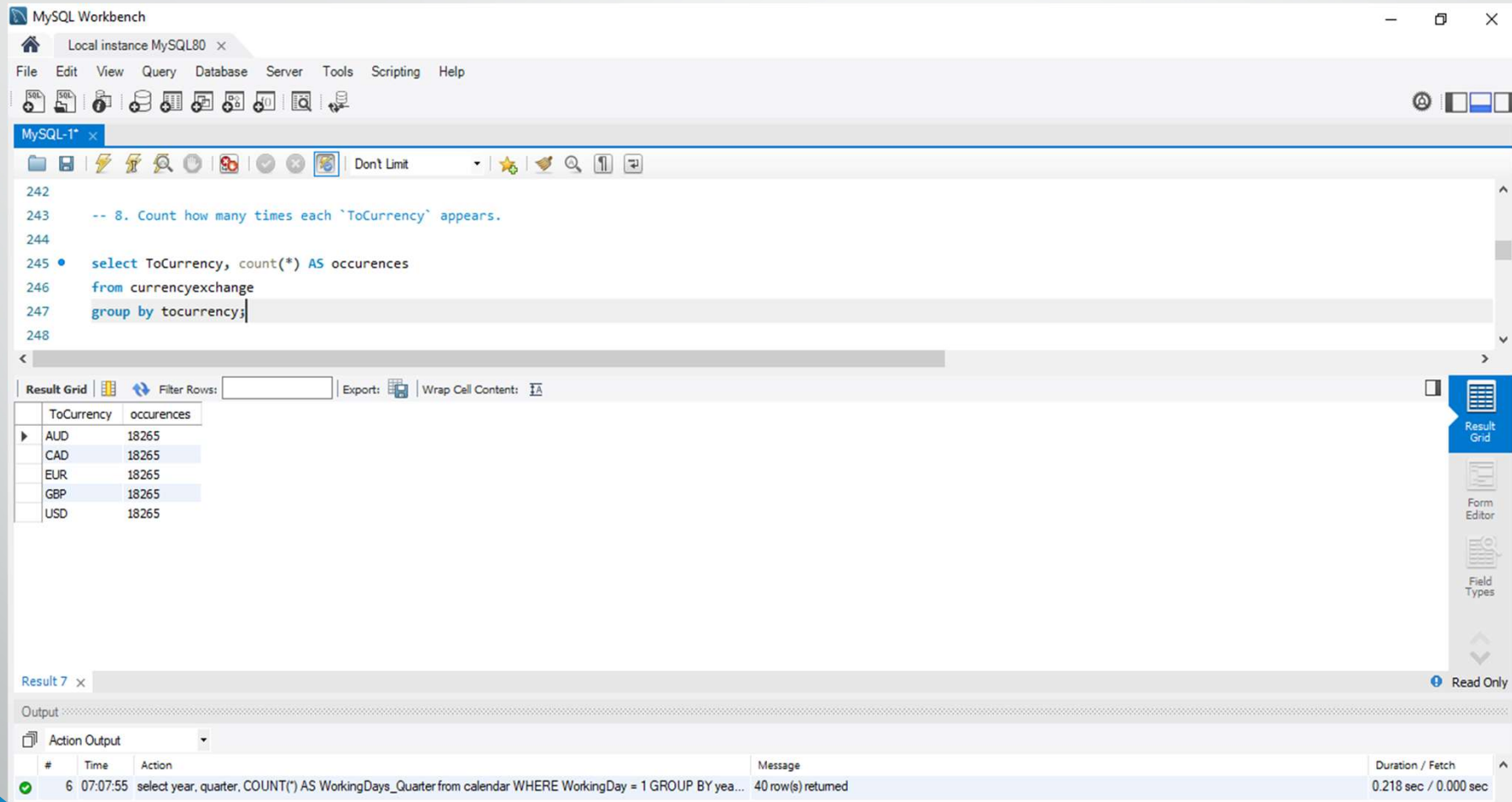
Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 5	07:06:46	select FromCurrency, MAX(exchange) AS Exchange_Max from currencyexchange GROUP BY FromCurrency	5 row(s) returned	0.937 sec / 0.000 sec

10. Count how many times each 'ToCurrency' appears.



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- 8. Count how many times each 'ToCurrency' appears.  
select ToCurrency, count(*) AS occurrences  
from currencyexchange  
group by tocurrency;
```

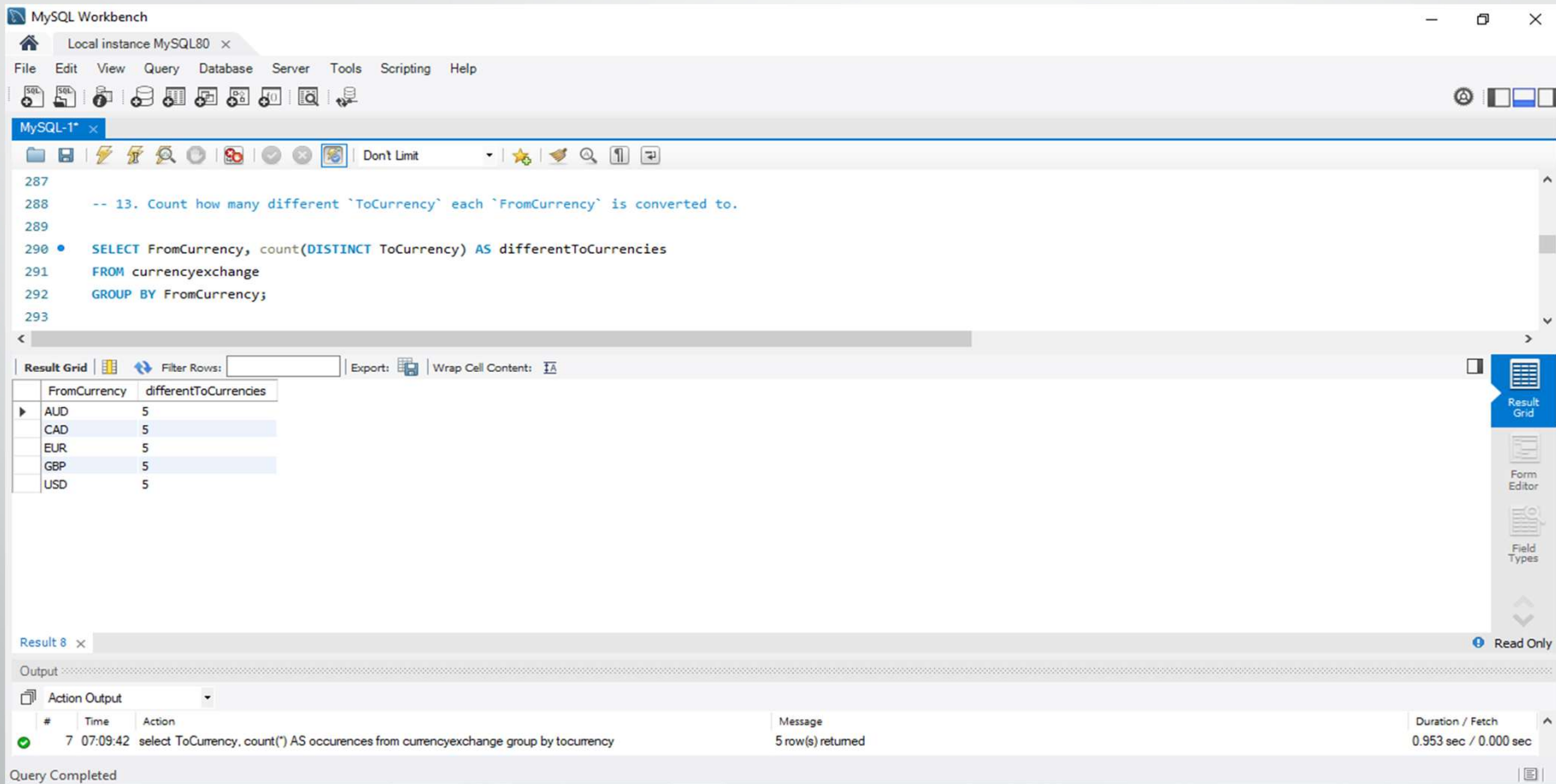
The query has been executed, and the results are displayed in the 'Result Grid' tab. The results show the count of occurrences for each 'ToCurrency'.

ToCurrency	occurrences
AUD	18265
CAD	18265
EUR	18265
GBP	18265
USD	18265

The 'Output' tab shows the execution details for 'Result 7':

#	Time	Action	Message	Duration / Fetch
6	07:07:55	select year, quarter, COUNT(*) AS WorkingDays_Quarter from calendar WHERE WorkingDay = 1 GROUP BY yea...	40 row(s) returned	0.218 sec / 0.000 sec

11. Count how many different `ToCurrency` each `FromCurrency` is converted to.



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- 13. Count how many different `ToCurrency` each `FromCurrency` is converted to.
SELECT FromCurrency, count(DISTINCT ToCurrency) AS differentToCurrencies
FROM currencyexchange
GROUP BY FromCurrency;
```

The query results are displayed in the Result Grid, showing 5 rows of data:

FromCurrency	differentToCurrencies
AUD	5
CAD	5
EUR	5
GBP	5
USD	5

The Output panel at the bottom shows the query execution details:

#	Time	Action	Message	Duration / Fetch
7	07:09:42	select ToCurrency, count(*) AS occurrences from currencyexchange group by tocurrency	5 row(s) returned	0.953 sec / 0.000 sec

Query Completed

12. Show 20 records starting from the 11th row.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

MySQL-1*

Don't Limit

```
-- 3. Show 20 records starting from the 11th row.  
SELECT *  
FROM calendar  
LIMIT 20 OFFSET 10;  
  
SELECT *  
FROM currencyexchange  
LIMIT 20 OFFSET 10;
```

Result Grid

Date	FromCurrency	ToCurrency	Exchange
2015-01-01	EUR	AUD	1.4829
2015-01-01	EUR	CAD	1.4063
2015-01-01	EUR	EUR	1
2015-01-01	EUR	GBP	0.7789
2015-01-01	EUR	USD	1.2141
2015-01-01	GBP	AUD	1.90384
2015-01-01	GBP	CAD	1.80549
2015-01-01	GBP	EUR	1.28386
2015-01-01	GBP	GBP	1
2015-01-01	GBP	USD	1.55874
2015-01-01	USD	AUD	1.2214

currencyexchange 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	07:11:39	SELECT FromCurrency, count(DISTINCT ToCurrency) AS differentToCurrencies FROM currencyexchange GROU...	5 row(s) returned	0.969 sec / 0.000 sec

Query Completed

13. How to get 15 non-working days.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
398
399
400 -- 6. Get 15 non-working days.
401 SELECT *
402 FROM calendar
403 WHERE WorkingDay= 0
404 LIMIT 15;
405
406
```

Result Grid

	date	datekey	year	qyear	YearQuarterNumber	quarter	yearmonth	yearmonthshort	YearMonthNumber	Month	MonthShort	MonthNumber	DayofWeek	DayofWeekShort	DayofWeekNumber	WorkingDay
	2015-01-01	20150101	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Thursday	Thu	5	0
	2015-01-03	20150103	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0
	2015-01-04	20150104	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0
	2015-01-10	20150110	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0
	2015-01-11	20150111	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0
	2015-01-17	20150117	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0
	2015-01-18	20150118	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0
	2015-01-19	20150119	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Monday	Mon	2	0
	2015-01-24	20150124	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0
	2015-01-25	20150125	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0

calendar 10 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	07:14:41	SELECT * FROM currencyexchange LIMIT 20 OFFSET 10	20 row(s) returned	0.141 sec / 0.000 sec

Query Completed

14. List the top 10 exchange rates from 'USD' to others.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1 x

Don't Limit

```
416
417 -- 8. List the top 10 exchange rates from 'USD' to others.
418
419 • SELECT *
420 FROM currencyexchange
421 WHERE FromCurrency = "USD"
422 ORDER BY Exchange DESC
423 LIMIT 10;
424
```

Result Grid

	Date	FromCurrency	ToCurrency	Exchange
▶	2020-03-19	USD	AUD	1.7253
	2020-03-23	USD	AUD	1.72364
	2020-03-20	USD	AUD	1.70225
	2020-03-22	USD	AUD	1.70225
	2020-03-21	USD	AUD	1.70225
	2020-03-18	USD	AUD	1.69819
	2020-03-24	USD	AUD	1.68911
	2020-04-03	USD	AUD	1.66936
	2020-04-05	USD	AUD	1.66936
	2020-04-04	USD	AUD	1.66936

currencyexchange 11 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 10	07:16:46	SELECT * FROM calendar WHERE WorkingDay= 0 LIMIT 15	15 row(s) returned	0.234 sec / 0.000 sec

Query Completed

15. List all products and their categories.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
569
570 -- 15. List all products and their categories.
571
572 • SELECT ProductName, CategoryName, SubCategoryName
573 FROM products;
574
```

The query is executed, and the results are displayed in the Result Grid. The results show 19 rows of product data.

ProductName	CategoryName	SubCategoryName
Contoso 512MB MP3 Player E51 Silver	Audio	MP4&MP3
Contoso 512MB MP3 Player E51 Blue	Audio	MP4&MP3
Contoso 1G MP3 Player E100 White	Audio	MP4&MP3
Contoso 2G MP3 Player E200 Silver	Audio	MP4&MP3
Contoso 2G MP3 Player E200 Red	Audio	MP4&MP3
Contoso 2G MP3 Player E200 Black	Audio	MP4&MP3
Contoso 2G MP3 Player E200 Blue	Audio	MP4&MP3
Contoso 4G MP3 Player E400 Silver	Audio	MP4&MP3
Contoso 4G MP3 Player E400 Black	Audio	MP4&MP3
Contoso 4G MP3 Player E400 Green	Audio	MP4&MP3
Contoso 4G MP3 Player E400 Orange	Audio	MP4&MP3
Contoso 4GB Flash MP3 Player E401 ...	Audio	MP4&MP3
Contoso 4GB Flash MP3 Player E401 ...	Audio	MP4&MP3
Contoso 4GB Flash MP3 Player E401 ...	Audio	MP4&MP3

The bottom of the interface shows the Output tab with the following information:

#	Time	Action	Message	Duration / Fetch
21	07:52:55	desc customer	24 row(s) returned	0.141 sec / 0.000 sec

16. Get customers who have never placed an order.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
615 -- Get customers who have never placed an order.
616 SELECT *
617 FROM customer c
618 WHERE c.CustomerKey NOT IN (
619     SELECT DISTINCT CustomerKey
620     FROM sales);
```

Result Grid

	CustomerKey	GeoAreaKey	StartDT	EndDT	Continent	Gender	Title	GivenName	MiddleInitial	Surname	StreetAddress	City	State	StateFull	ZipCode	Country	CountryFull	Birth
▶	210	2	1980-09-28	2030-05-08	Australia	female	Mrs.	Natalie	L	Hilder	63 Davenport Street	PADDYS FLAT	NSW	New South Wales	2632	AU	Australia	1950-
	322	2	1993-09-27	2031-06-30	Australia	female	Ms.	Alexandra	D	Stephen	61 Hart Street	GREEN CREEK	NSW	New South Wales	2338	AU	Australia	1974-
	609	7	1987-09-16	2007-10-13	Australia	female	Mrs.	Lucy	K	Parrott	22 Thomas Lane	BELLFIELD	VIC	Victoria	3081	AU	Australia	1947-
	661	2	1985-02-17	2007-11-19	Australia	male	Mr.	Archie	J	Hugh	60 Gaggin Street	KARUAH	NSW	New South Wales	2324	AU	Australia	1994-
	827	2	1983-06-07	2042-12-16	Australia	female	Ms.	Erin	B	Kellermann	48 Elgin Street	OAKHAMPTON	NSW	New South Wales	2320	AU	Australia	1980-
	855	7	1987-08-03	2018-08-02	Australia	female	Ms.	Kaitlyn	B	Bogle	10 Edmundsons Road	MOLLONGHIP	VIC	Victoria	3352	AU	Australia	1999-
	865	8	1984-06-10	2038-02-06	Australia	male	Mr.	Zane	A	Shellshear	43 Lapko Road	TINGLEDALE	WA	Western Australia	6333	AU	Australia	1978-
	928	8	1985-12-08	2009-12-18	Australia	female	Mrs.	Caitlyn	M	Viner	69 Alfred Street	BANDY CREEK	WA	Western Australia	6450	AU	Australia	1937-
	977	4	1988-03-16	2038-05-03	Australia	female	Dr.	Mariam	J	Bourke	59 Ross Street	MOUNT NATHAN	QLD	Queensland	4211	AU	Australia	1968-
	1259	7	1980-04-03	2013-08-18	Australia	male	Mr.	William	E	Fairfax	54 Gaffney Street	NORLANE	VIC	Victoria	3214	AU	Australia	1963-
	1291	5	1986-01-31	2012-08-20	Australia	female	Ms.	Ellie	O	Cayley	11 Gadd Avenue	KALLORA	SA	South Australia	5550	AU	Australia	1990-
	1306	6	1991-08-13	2008-04-12	Australia	female	Ms.	Ruby	I	Michell	99 Pelican Road	TOLMANS HILL	TAS	Tasmania	7007	AU	Australia	1989-
	1442	8	1989-08-02	2017-05-08	Australia	male	Mr.	Daniel	A	Pelsaert	95 Jacolite Street	SAWYERS VALLEY	WA	Western Australia	6074	AU	Australia	1952-

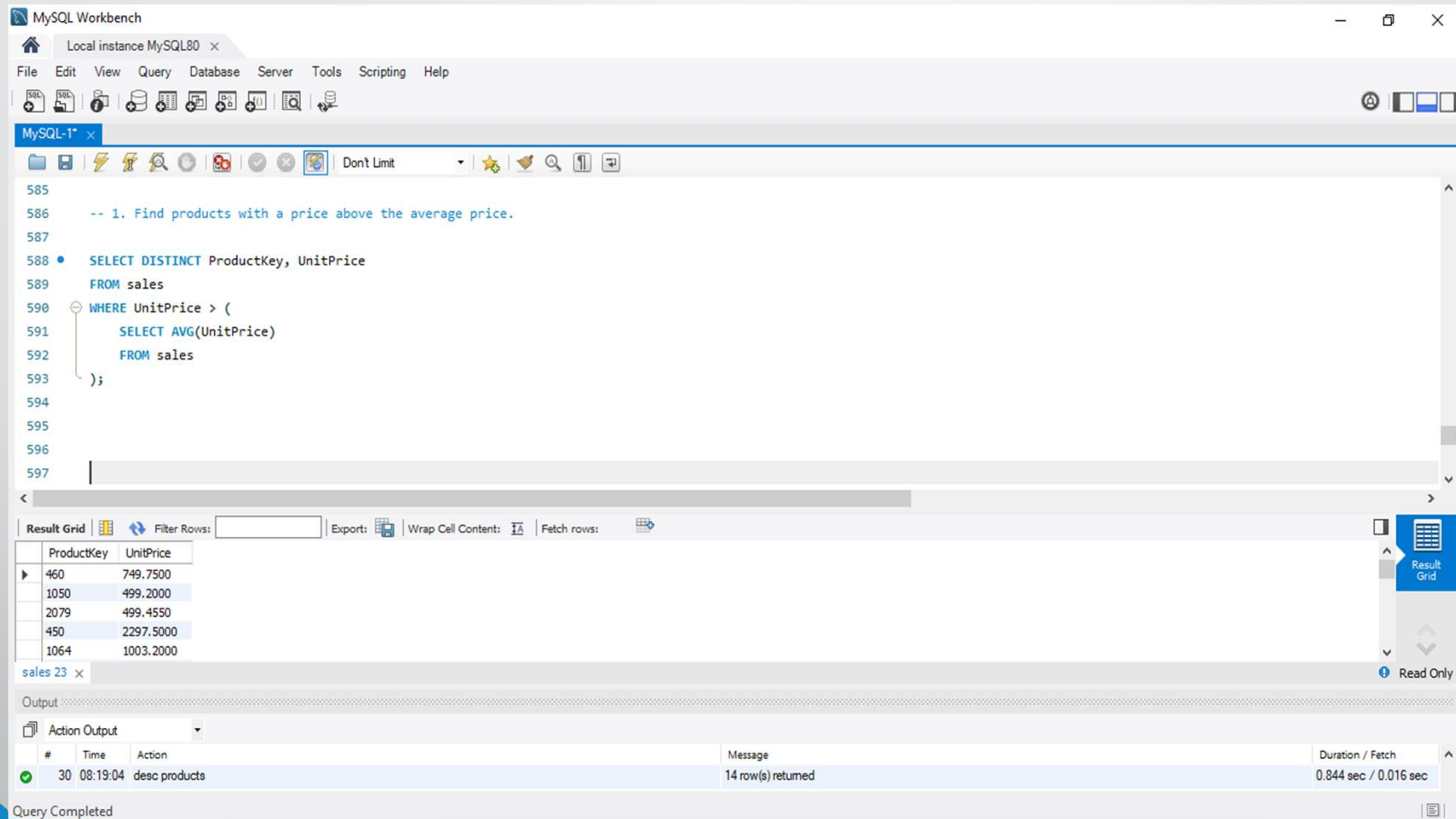
customer 21 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 28	08:15:15	SELECT * FROM customer c WHERE c.CustomerKey NOT IN (SELECT DISTINCT CustomerKey FROM sal...	4503 row(s) returned	23.594 sec / 0.750 sec

17. Find products with a price above the average price.



The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
-- 1. Find products with a price above the average price.  
  
SELECT DISTINCT ProductKey, UnitPrice  
FROM sales  
WHERE UnitPrice > (  
    SELECT AVG(UnitPrice)  
    FROM sales  
);
```

The results are displayed in the Result Grid, showing 14 rows of data:

ProductKey	UnitPrice
460	749.7500
1050	499.2000
2079	499.4550
450	2297.5000
1064	1003.2000

The status bar at the bottom indicates "Query Completed" and "14 row(s) returned".

18. List all orders with the customer's name included.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
528
529 -- List all orders with the customer's name included.
530 • SELECT s.OrderKey, s.OrderDate, s.ProductKey, s.Quantity, s.NetPrice,
531      c.GivenName
532 FROM sales s
533 JOIN customer c
534   ON s.CustomerKey = c.CustomerKey;
535
536
537
538
539
540
```

Result Grid

	OrderKey	OrderDate	ProductKey	Quantity	NetPrice	GivenName
▶	10080	2015-01-01	10	2	40.4933	Toby
	10080	2015-01-01	1793	5	26.4880	Toby
	10080	2015-01-01	1686	1	4.4526	Toby
	10080	2015-01-01	1274	2	83.4080	Toby
	10080	2015-01-01	1834	1	1213.5150	Toby

Result 24 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 31	08:19:29	SELECT DISTINCT ProductKey, UnitPrice FROM sales WHERE UnitPrice > (SELECT AVG(UnitPrice) FRO...	3677 row(s) returned	15.110 sec / 0.016 sec

19. How to retrieve a list of customers and their most recent order date .

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x SQL File 4*

Don't Limit

```
668 -- Retrieve a list of customers and their most recent order date
669
670 • SELECT s.orderkey, s.orderdate, s.productkey, s.quantity, s.netprice, c.givenname
671 FROM sales s
672 Join customer as c
673 on s.CustomerKey = c.CustomerKey;
674
```

Result Grid

	orderkey	orderdate	productkey	quantity	netprice	givenname
▶	10080	2015-01-01	10	2	40.4933	Toby
	10080	2015-01-01	1793	5	26.4880	Toby
	10080	2015-01-01	1686	1	4.4526	Toby
	10080	2015-01-01	1274	2	83.4080	Toby
	10080	2015-01-01	1834	1	1213.5150	Toby
	10080	2015-01-01	1645	1	30.2134	Toby
	20014	2015-01-02	139	4	399.9920	Callum
	20014	2015-01-02	1271	5	11.1200	Callum
	20014	2015-01-02	104	4	75.9000	Callum
	20059	2015-01-02	1609	2	137.2747	Dylan
	30053	2015-01-03	262	3	148.0560	Dean
	30053	2015-01-03	106	3	87.7734	Dean
	30090	2015-01-03	825	2	34.7500	Cody
	30105	2015-01-03	1426	4	1060.2000	Molly

Result 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 5	06:47:35	SELECT s.orderkey, s.orderdate, s.productkey, s.quantity, s.netprice, c.givenname FROM sales s Join customer as...	420200 row(s) returned	0.125 sec / 5.500 sec

20. How to limit exchange rates to the first 12 EUR-USD conversions.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

MySQL-1* x

Don't Limit

```
-- 19. Limit exchange rates to the first 12 EUR-USD conversions.
504 SELECT *
505 FROM currencyexchange
506 WHERE FromCurrency= "EUR" AND ToCurrency= "USD"
507 ORDER BY Date ASC
508 LIMIT 12;
```

Result Grid

	Date	FromCurrency	ToCurrency	Exchange
▶	2015-01-01	EUR	USD	1.2141
	2015-01-02	EUR	USD	1.2043
	2015-01-03	EUR	USD	1.2043
	2015-01-04	EUR	USD	1.2043
	2015-01-05	EUR	USD	1.1915
	2015-01-06	EUR	USD	1.1914
	2015-01-07	EUR	USD	1.1831
	2015-01-08	EUR	USD	1.1768
	2015-01-09	EUR	USD	1.1813
	2015-01-10	EUR	USD	1.1813
	2015-01-11	EUR	USD	1.1813
	2015-01-12	EUR	USD	1.1804

currencyexchange 28 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
35	08:29:03	select * from calendar ORDER BY DayofWeekNumber, date ASC LIMIT 10	10 row(s) returned	0.250 sec / 0.016 sec

Query Completed

CONCLUSION

- The Contoso Retail Database provides a well-structured sample dataset that represents a real-world retail business. It contains detailed information about sales transactions, products, customers, stores, and time, making it ideal for practicing SQL queries, data analysis, and business intelligence.
- By analyzing this data, we can:
- Understand sales trends over time
- Identify best-selling products and top-performing stores
- Explore customer demographics and buying patterns
- Measure profitability and business performance
- This dataset is especially useful for students and analysts to build practical skills in tools like Power BI, SQL, and Excel, and to simulate real business scenarios for decision-making.

THANK YOU

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